

SAMADH HIGHER SECONDARY SCHOOL — TRICHY-20

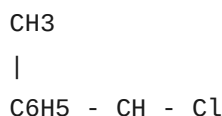
Class XII Chemistry | Complete Answer Key with Structures & Reactions

Quarterly Exam 2026-27 | 70 Marks | NEP 2020 NCERT Edition (10 Chapters)

NEP 2020 Chapters: Ch.1 Solutions | Ch.6 Haloalkanes & Haloarenes | Ch.7 Alcohols, Phenols & Ethers | Ch.9 Amines | Ch.10 Biomolecules (Part II page numbers from 2025-26 reprint used throughout)

Q1. Answer: C) 1-Chloro-1-phenylethane

Structure of compound given:



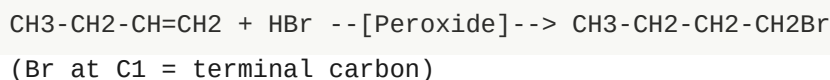
Longest chain = 2C (ethane). Cl and phenyl both on C1.

IUPAC Name: 1-chloro-1-phenylethane

NCERT (NEP 2020): Ch.6, p.174 — IUPAC nomenclature of haloalkanes

Q2. Answer: A) 1-Bromobutane [anti-Markovnikov product]

Reaction (Peroxide = anti-Markovnikov):

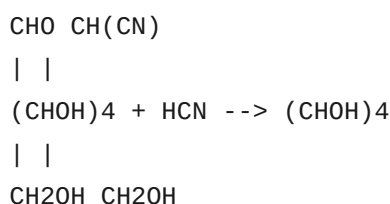


Without peroxide: Br adds to C2 (Markovnikov). With peroxide: Br adds to C1 (anti-Markovnikov).

NCERT (NEP 2020): Ch.6, p.181 — Anti-Markovnikov addition (peroxide effect)

Q3. Answer: B) Cyanohydrin of glucose

Reaction of glucose with HCN:



HCN adds to -CHO (aldehyde) group at C1. -CN attaches to carbonyl carbon.

Answer B matches this structure.

NCERT (NEP 2020): Ch.10, p.293 — Chemical reactions of glucose

Q4. Answer: B) Increased blood clotting time

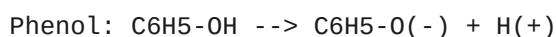
Vitamin K is essential for synthesis of prothrombin and clotting factors.

Deficiency of Vit-K --> impaired blood clotting --> increased clotting time.

NCERT (NEP 2020): Ch.10, p.317-318 — Vitamins table

Q5. Answer: C) Resonance stabilization of phenoxide ion

Why phenol is more acidic than ethanol:



(phenoxide ion – stabilised by resonance with benzene ring)

Ethanol: $\text{C}_2\text{H}_5\text{-OH} \rightarrow \text{C}_2\text{H}_5\text{-O}(-) + \text{H}(+)$
(ethoxide ion – NO resonance stabilisation)

Phenoxide is more stable $\rightarrow$ phenol loses H^+ more easily $\rightarrow$ more acidic.

NCERT (NEP 2020): Ch.7, p.213 — Acidity of phenols

Q6. Answer: D) p-nitrophenol (strongest acid)

Effect of $-\text{NO}_2$ (EWG) at para position:

OH OH

| |

(benzene) (benzene)

| |

(no sub) NO_2 (para)

$\text{pK}_a \sim 10.0$ $\text{pK}_a \sim 7.1$ (more acidic)

$-\text{NO}_2$ withdraws electrons from ring by resonance $\rightarrow$ stabilises phenoxide ion more strongly.

More stable phenoxide = stronger acid. Order: p-nitrophenol > p-cresol > o-cresol > phenol

NCERT (NEP 2020): Ch.7, p.213-214 — Effect of substituents on acidity of phenols

Q7. Answer: B) Positive (+ve) deviation from Raoult's law

Ethanol + water azeotrope:

Ethanol-water A-B H-bonds are WEAKER than ethanol-ethanol and water-water H-bonds.

Molecules escape more easily $\rightarrow$ vapour pressure HIGHER than expected (positive deviation).

$p_{\text{observed}} > x_1.p^*_1 + x_2.p^*_2$ (positive deviation)

Boiling point of azeotrope is LESS than both pure components.

NCERT (NEP 2020): Ch.1, p.8-11 — Non-ideal solutions, positive deviation

Q8. Answer: D) 2M K_2SO_4 (highest boiling point)

Van't Hoff factor comparison:

1M $\text{KCl} \rightarrow \text{K}^+ + \text{Cl}^-$ $i = 2$, effective = $1 \times 2 = 2$

2M $\text{KCl} \rightarrow \text{K}^+ + \text{Cl}^-$ $i = 2$, effective = $2 \times 2 = 4$

1M $\text{K}_2\text{SO}_4 \rightarrow 2\text{K}^+ + \text{SO}_4^{2-}$ $i = 3$, effective = $1 \times 3 = 3$

2M $\text{K}_2\text{SO}_4 \rightarrow 2\text{K}^+ + \text{SO}_4^{2-}$ $i = 3$, effective = $2 \times 3 = 6$ <-- HIGHEST

More particles $\rightarrow$ greater boiling point elevation. Answer: 2M K_2SO_4 .

NCERT (NEP 2020): Ch.1, p.24-27 — Van't Hoff factor, elevation of boiling point

Q9. Answer: B) NaCl dissociates into ions

Osmotic pressure comparison:

$\text{NaCl} (0.1\text{M}) \rightarrow \text{Na}^+ + \text{Cl}^-$ $i = 2 \rightarrow \pi = 2 \times C \times R \times T$

Glucose (0.1M) $\rightarrow$ no dissociation $i = 1 \rightarrow \pi = 1 \times C \times R \times T$

NaCl has twice as many particles $\rightarrow$ twice the osmotic pressure.

NCERT (NEP 2020): Ch.1, p.19-21 — Osmotic pressure; Van't Hoff factor

Q10. Answer: B) Decreases in vapour pressure

Non-volatile solute lowers vapour pressure of solvent (Raoult's law).

$p_{\text{solution}} = x_{\text{solvent}} \times p^*_{\text{solvent}}$ (lower than pure solvent)

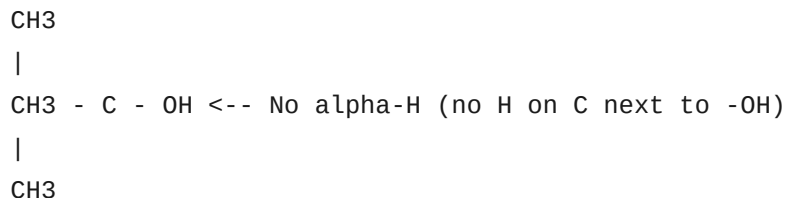
For freezing: vapour pressure of solution must equal vapour pressure of pure solid.

Since solution has LOWER vapour pressure, this happens at a LOWER temperature.
Therefore freezing point is DEPRESSED.

NCERT (NEP 2020): Ch.1, p.15-17 — Depression in freezing point

Q11. Answer: C) Tertiary Alcohol

Structure showing no alpha-H in tertiary alcohol:

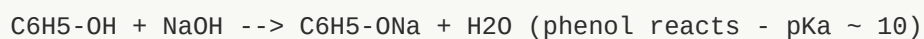


No alpha-H --> cannot be oxidised by KMnO_4 / $\text{K}_2\text{Cr}_2\text{O}_7$.

On heating with conc. H_2SO_4 --> gives alkene by dehydration.

NCERT (NEP 2020): Ch.7, p.232 — Oxidation of alcohols

Q12. Answer: C) Phenol is more Acidic



Phenol is acidic enough to react with NaOH. Ethanol is not.

NCERT (NEP 2020): Ch.7, p.213 — Reactions of phenol with NaOH

Q13. Answer: A) Both A and R true; R correctly explains A

A: Lucas test distinguishes 1 deg, 2 deg, 3 deg alcohols. (TRUE)

R: 3 deg alcohols form carbocations more easily than others. (TRUE)

R explains A: Because 3 deg carbocation forms instantly, turbidity appears immediately.

NCERT (NEP 2020): Ch.7, p.228 — Lucas test

Q14. Answer: B) Both true; R is NOT the correct explanation of A

A: Glycogen is stored in animals (not starch). (TRUE)

R: Glycogen is more highly branched than starch. (TRUE)

But branching is NOT why animals store glycogen — the reason is faster glucose mobilisation.

R does not explain A, so answer is B.

NCERT (NEP 2020): Ch.10, p.300-301 — Polysaccharides: glycogen and starch

Q15. Answer: A) Both true; R correctly explains A

A: Vitamins essential in small amounts. (TRUE)

R: Body cannot synthesise adequate amounts. (TRUE)

R is the direct reason why we must get vitamins through diet. Correct explanation.

NCERT (NEP 2020): Ch.10, p.317 — Introduction to vitamins

Q16. Answer: C) A is true, R is false

A: Denaturation leads to loss of biological activity. (TRUE — 3D shape is lost)

R: FALSE. Denaturation disrupts 2 deg and 3 deg structures only.

Primary structure (peptide bonds) remains INTACT during denaturation.

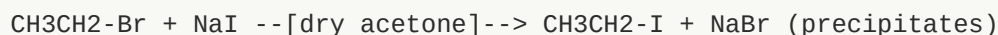
Primary structure = covalent peptide bonds (-CO-NH-) = NOT broken by denaturation

2 deg / 3 deg = H-bonds, disulphide bonds, hydrophobic interactions = BROKEN

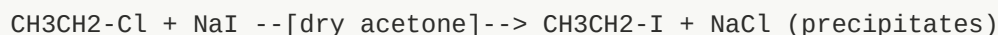
NCERT (NEP 2020): Ch.10, p.311 — Denaturation of proteins

Q17. Finkelstein reaction and Swarts reaction

(i) FINKELSTEIN REACTION:



~or:

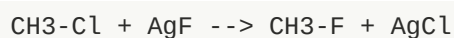


NaBr / NaCl are insoluble in dry acetone and PRECIPITATE OUT.

This drives equilibrium forward (Le Chatelier's principle).

Use: To prepare alkyl iodides from alkyl chlorides/bromides.

(ii) SWARTS REACTION:



Alkyl halide + metallic fluoride (AgF or Hg₂F₂) → alkyl fluoride.

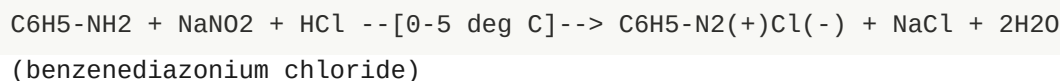
Direct fluorination is too violent — Swarts reaction is the controlled method.

NCERT (NEP 2020): Ch.6, p.178-179 — Finkelstein reaction and Swarts reaction

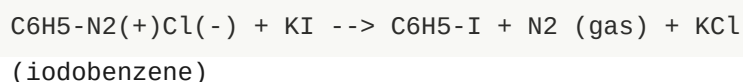
Q18. Convert (i) Aniline to Iodobenzene (ii) Chloromethane to Ethane

(i) ANILINE → IODOBENZENE (Diazotisation + Sandmeyer):

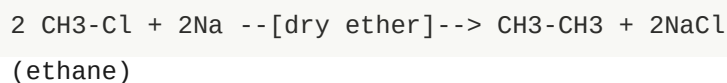
Step 1 — Diazotisation:



Step 2 — Sandmeyer reaction:



(ii) CHLOROMETHANE → ETHANE (Wurtz Reaction):



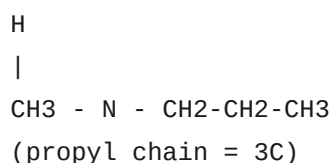
Na metal couples two alkyl halide molecules. Gives alkane with double carbons.

NCERT (NEP 2020): (i) Ch.9, p.279-280 — Diazonium salts, Sandmeyer | (ii) Ch.6, p.177 — Wurtz reaction

Q19. IUPAC Names

(i) CH₃-N(H)-CH₂CH₂CH₃:

Full structure:



Longest chain from N: propan-1-amine (3C). CH₃ is N-substituent.

IUPAC Name: N-methylpropan-1-amine

(ii) Cyclohexyl-N(CH₃)₂:

Full structure:

(cyclohexane ring) - N(CH₃)₂

(two methyl groups on N)

Base name: cyclohexanamine. Two methyls on N = N,N-dimethyl.

!IUPAC Name: N,N-dimethylcyclohexanamine

NCERT (NEP 2020): Ch.9, p.260-261 — IUPAC nomenclature of amines

Q20. Gabriel Phthalimide Synthesis

FULL REACTION SEQUENCE WITH STRUCTURES:

Step 1 — Form potassium salt:

Phthalimide + KOH --> Potassium phthalimide + H₂O

(C₈H₅NO₂) (C₈H₄NO₂(-) K(+))

(has N-H bond) (N-H replaced by N-K)

Step 2 — N-alkylation (S_N2):

K-phthalimide + R-X --> N-alkyl phthalimide + KX

(nucleophile) (alkyl (R now on N)

halide)

Example: R-X = CH₃Br gives N-methylphthalimide

Step 3 — Hydrolysis (acidic or basic):

N-alkylphthalimide + NaOH/H₂O --> R-NH₂ + sodium phthalate

e.g. N-methylphthalimide --> CH₃NH₂ (methylamine = primary amine)

KEY POINTS:

-> Gives ONLY primary amine (1 deg) — no 2 deg or 3 deg contamination.

-> CANNOT make aromatic amines (aryl halides won't do S_N2).

NCERT (NEP 2020): Ch.9, p.264-265 — Gabriel phthalimide synthesis

Q21(A). DNA: 30% Adenine — Calculate % Guanine [OR: Vitamin deficiency diseases]

CHARGAFF'S RULE: In DNA, A=T and G=C

Given: A = 30%

Step 1: T = 30% (because A = T)

Step 2: A + T = 60%

Step 3: G + C = 100 - 60 = 40%

Step 4: G = 40/2 = 20% (because G = C)

ANSWER: Guanine = 20%

OR — Vitamin deficiency diseases:

Vitamin D --> Rickets (children), Osteomalacia (adults)

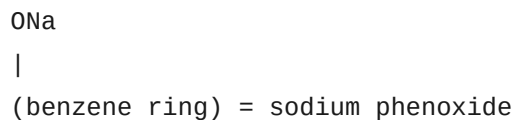
Vitamin B₁₂ --> Pernicious anaemia (lack of RBCs)

NCERT (NEP 2020): DNA: Ch.10, p.312-314 — Nucleic acids, Chargaff's rule | Vitamins: Ch.10, p.317-320

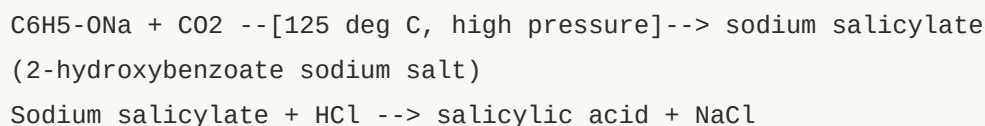
Q22. Kolbe's reaction and Reimer-Tiemann reaction

(i) KOLBE'S REACTION:

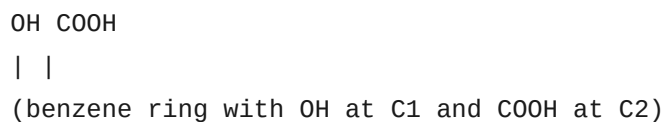
Reactant — Sodium phenoxide:



Reaction:



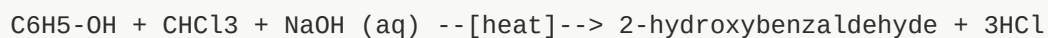
Product — Salicylic acid (2-hydroxybenzoic acid):



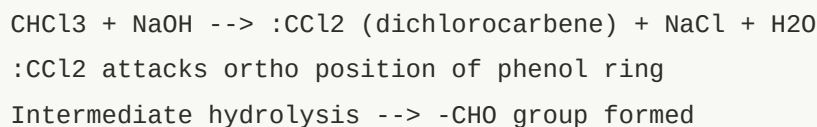
Use: Salicylic acid is used to make aspirin (acetylsalicylic acid).

(ii) REIMER-TIEMANN REACTION:

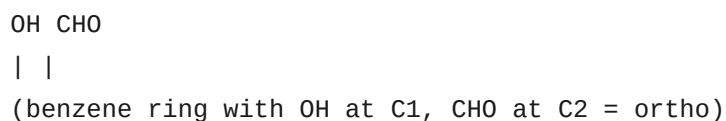
Reaction:



Mechanism — intermediate:



Product — Salicylaldehyde (2-hydroxybenzaldehyde):



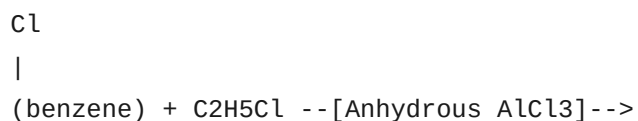
Major product: ortho (salicylaldehyde). Minor: para-hydroxybenzaldehyde.

NCERT (NEP 2020): Ch.7, p.226-227 — Kolbe's and Reimer-Tiemann reactions of phenol

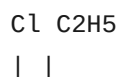
Q23. Complete reactions (three parts)

(i) Friedel-Crafts Alkylation of chlorobenzene:

Reactants:



Product (ethyl group enters ortho/para to Cl):

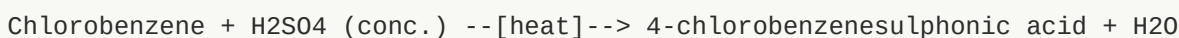


(benzene: Cl at C1, C2H5 at C4 = para = major product)



AlCl₃ generates C₂H₅(+) carbocation (electrophile). -Cl is ortho/para director.

(ii) Sulphonation of chlorobenzene:



(benzene: Cl at C1, SO₃H at C4 = para = major product)

-Cl directs SO₃H to para position.

(iii) CH₃OH + SOCl₂:



Both byproducts are gases that escape → pure CH₃Cl obtained.

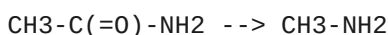
Best method for converting alcohol to alkyl chloride.

NCERT (NEP 2020): (i)(ii) Ch.6, p.198-201 — Reactions of haloarenes | (iii) Ch.7, p.228 — SOCl₂ with alcohols

Q24. Construct reactions (three conversions)

(i) CH₃CONH₂ → CH₃NH₂ (Hofmann Bromamide Degradation):

Structure of reactant (acetamide) and product (methylamine):

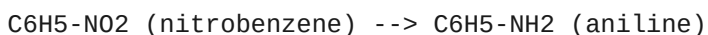


(acetamide, 2C) (methylamine, 1C)



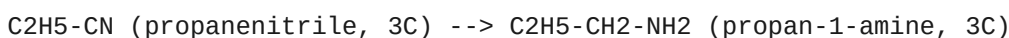
Product amine has ONE CARBON LESS than starting amide.

(ii) C₆H₅NO₂ → C₆H₅NH₂ (Reduction of nitrobenzene):



Industrial method: Fe/HCl. Lab: Sn/HCl or H₂/Ni.

(iii) C₂H₅CN → C₃H₇NH₂ (Reduction of nitrile):



Nitrile gives primary amine with SAME carbon count.

NCERT (NEP 2020): Ch.9, p.264-268 — Hofmann bromamide (p.266), Reduction of nitro (p.264), Reduction of nitrile (p.265)

Q25. Numerical — Mass of ascorbic acid (Kf problem)

!Given data:

$$\Delta T_f = 1.5 \text{ K} \quad | \quad K_f (\text{acetic acid}) = 2.93 \text{ K kg/mol} \quad | \quad \text{solvent mass} = 75 \text{ g} = 0.075 \text{ kg}$$

!Molar mass of ascorbic acid C₆H₈O₆:

$$\text{C: } 6 \times 12 = 72$$

$$\text{H: } 8 \times 1 = 8$$

$$\text{O: } 6 \times 16 = 96$$

$$\text{Total } M_2 = 176 \text{ g/mol}$$

!Formula: $\Delta T_f = K_f \times (w_2 / M_2) / w_1(\text{kg})$

$$1.5 = 2.93 \times (w_2 / 176) / 0.075$$

$$1.5 = 2.93 \times w_2 / (176 \times 0.075)$$

$$1.5 = 2.93 \times w_2 / 13.2$$

$$w_2 = (1.5 \times 13.2) / 2.93$$

$$w_2 = 19.8 / 2.93$$

$$w_2 = 6.76 \text{ g}$$

ANSWER: 6.76 g of ascorbic acid must be dissolved.

NCERT (NEP 2020): Ch.1, p.15-17 — Depression in freezing point; solved examples p.20-23

Q26. Mixtures A (Benzene+Toluene) and B (Ethanol+Water)

Q1: Which is non-ideal?

Mixture B (Ethanol + Water) is NON-IDEAL — NEGATIVE deviation.

Mixture A (Benzene + Toluene) is IDEAL — obeys Raoult's law.

Q2: Reason for negative deviation in Mixture B:

Intermolecular forces comparison:

Ethanol alone: $\text{C}_2\text{H}_5\text{-OH} \cdots \text{HO-C}_2\text{H}_5$ (ethanol-ethanol H-bond)

Water alone: $\text{H-O-H} \cdots \text{H-O-H}$ (water-water H-bond)

Mixed solution: $\text{C}_2\text{H}_5\text{-OH} \cdots \text{H-O-H}$ (ethanol-water H-bond = STRONGER)

A-B interactions are STRONGER than A-A or B-B.

Molecules are held more tightly --> escape tendency decreases.

$$p_{\text{observed}} < x_1.p^*_1 + x_2.p^*_2 \text{ (negative deviation: lower vapour pressure)}$$

$\Delta H_{\text{mix}} < 0$ (heat released), $\Delta V_{\text{mix}} < 0$ (volume contracts)

Q3: Raoult's Law:

For a solution of volatile liquids, partial vapour pressure of each component is proportional to its mole fraction:

$$p_1 = x_1 \times p^*_1 \quad p_2 = x_2 \times p^*_2$$

$$\text{Total } P = x_1.p^*_1 + x_2.p^*_2$$

NCERT (NEP 2020): Ch.1, p.8-11 — Raoult's law; ideal and non-ideal solutions with deviations

Q27. Boiling point arrangements and solubility

(i) Increasing bp of alcohols:

Why alcohols have high bp — Intermolecular H-bonding:

$\text{R-O-H} \cdots \text{O-R}$ (H-bond between -OH groups)

H

Methanol(65C) < Ethanol(78C) < Propanol(97C) < Butan-2-ol(99C) <

Butan-1-ol(118C) < Pentanol(138C)

Longer chain = more van der Waals + H-bonding = higher bp.

Butan-2-ol < Butan-1-ol: branching in butan-2-ol reduces surface area.

(ii) bp order: Butane, Ethoxyethane, Pentanol:

Butane(-0.5C) < Ethoxyethane(35C) < Pentan-1-ol(138C)

Butane: only weak van der Waals (no dipole, no H-bond)

Diethyl ether: C=O dipole present, but NO O-H bond --> no self H-bonding

Pentanol: O-H group --> strong intermolecular H-bonds --> highest bp

(iii) Ethanol soluble but hexanol not in water:

Ethanol (C₂H₅OH):

Small molecule. -OH group forms H-bonds with water:

C₂H₅-O-H --- H-O-H (H-bond with water)

Hydrophilic -OH dominates over short C₂ chain --> MISCIBLE with water.

Hexanol (C₆H₁₃OH):

Long non-polar C₅ chain: CH₃-CH₂-CH₂-CH₂-CH₂-CH₂-OH

Hydrophobic C₅ chain dominates --> hydrophobic effect overcomes -OH H-bonding.

INSOLUBLE in water.

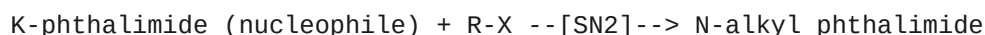
Rule: As carbon chain > 4C, alcohol water solubility decreases sharply.

NCERT (NEP 2020): Ch.7, p.210-212 — Physical properties of alcohols: boiling points and solubility

Q28. Gabriel synthesis cannot make aniline [OR: Henry's Law]

(i) Why Gabriel phthalimide CANNOT make aniline:

Gabriel requires SN2 reaction with alkyl halide:



For aniline (C₆H₅-NH₂), we would need chlorobenzene (C₆H₅-Cl) as R-X.

But chlorobenzene CANNOT undergo SN2 because:

In C₆H₅-Cl:

- Lone pair on Cl is in resonance with benzene pi system
- C-Cl bond has partial double bond character (stronger, shorter bond)
- Benzene pi cloud REPELS incoming nucleophile
- Therefore SN2 does NOT occur with aryl halides

So aniline CANNOT be made by Gabriel synthesis.

(ii) Why only 1 degree aliphatic amines:

Phthalimide N-H undergoes only monoalkylation (one R group enters N).

After hydrolysis: only R-NH₂ (1 deg amine). No R₂NH or R₃N contamination.

(iii) Alternative method for aniline:



(nitrobenzene) (aniline)

OR — HENRY'S LAW:

Statement: $p = K_H \times x$

p = partial pressure of gas above solution

K_H = Henry's law constant (different for each gas-solvent pair)

x = mole fraction of gas dissolved in solution

Higher K_H --> less soluble the gas.

Applications:

1. Soda drinks: CO₂ dissolved under high pressure. Escapes when bottle opened (K_H large).
2. Scuba diving: High pressure dissolves N₂ in blood. Rapid ascent = N₂ bubbles (bends).
He-O₂ mixture used (He has very high K_H = less soluble = safer).
3. Altitude: Low O₂ partial pressure --> less O₂ dissolves in blood --> altitude sickness.

NCERT (NEP 2020): Gabriel: Ch.9, p.264-265 | Henry's law: Ch.1, p.5-8

Q29. RBCs in three solutions — Osmosis

1) Identify solution types:

Solution A — cells swell and BURST:

Type: HYPOTONIC (solution less concentrated than cell contents)

Water moves IN to cells (osmosis) --> cells swell --> burst (haemolysis)

Solution B — cells stay NORMAL:

Type: ISOTONIC (same concentration as cell contents = 0.9% NaCl)

No net water movement --> cells stay normal

Solution C — cells SHRINK:

Type: HYPERTONIC (solution more concentrated than cell contents)

Water moves OUT of cells (osmosis) --> cells shrink (crenation / plasmolysis)

2) % NaCl for patients:

0.9% w/v NaCl = Normal Saline = ISOTONIC with blood plasma

Administered in IV drips. Does not damage RBCs.

3) Two real-life examples:

Example 1 — Wilted plant in water:

Plant cells are hypertonic relative to water --> water enters by osmosis --> plant regains turgidity.

Example 2 — Salt/sugar preservation:

Concentrated salt (pickle) or sugar (jam) creates hypertonic environment.

Bacteria lose water by plasmolysis --> die --> food is preserved.

NCERT (NEP 2020): Ch.1, p.19-22 — Osmotic pressure, isotonic solutions, real life examples

Q30. Biomolecules — 5-OH test, Anomers, Protein bonds, Zwitter ion

1) Test to confirm 5 -OH groups in glucose:

Glucose + excess $(\text{CH}_3\text{CO})_2\text{O}$ (acetic anhydride) --> Glucose pentaacetate

Five acetyl ($-\text{COCH}_3$) groups are incorporated = 5 free -OH groups confirmed.

Also: Glucose reacts with 5 moles of HIO_4 (periodic acid oxidation).

2) Anomers — Definition with example:

Anomers are cyclic sugar stereoisomers differing ONLY at C1 (anomeric carbon).

alpha-D-glucose vs beta-D-glucose:

alpha-D-glucose: -OH at C1 is on SAME side as C6 (axial) --> $[\alpha]_D = +112^\circ$

beta-D-glucose: -OH at C1 is on OPPOSITE side (equatorial) --> $[\alpha]_D = +18.7^\circ$

In solution, both interconvert (mutarotation) --> equilibrium $[\alpha]_D = +52.5^\circ$

3) Bonds at each protein structure level:

Primary structure: Covalent PEPTIDE BONDS ($-\text{CO}-\text{NH}-$) between amino acids

Secondary structure: HYDROGEN BONDS between $\text{C}=\text{O}$ and $\text{N}-\text{H}$ of backbone

alpha-helix: H-bonds within same chain (coil shape)

beta-pleated sheet: H-bonds between parallel/antiparallel chains

Tertiary structure: DISULPHIDE BONDS ($-\text{S}-\text{S}-$) between cysteine residues

+ HYDROGEN BONDS

+ IONIC BONDS (salt bridges between $-\text{NH}_3(+)$ and $-\text{COO}(-)$)

+ HYDROPHOBIC interactions (non-polar side chains)

4) Zwitter Ion:

Amino acid in water exists as DIPOLAR ION (zwitter ion):

Free form: $\text{H}_2\text{N} - \text{CHR} - \text{COOH}$ (uncharged)

Zwitter ion: $(+)\text{H}_3\text{N} - \text{CHR} - \text{COO}(-)$ (dipolar / internal salt)

-NH₂ accepts H⁺ and becomes -NH₃(+).

-COOH loses H⁺ and becomes -COO(-).

Overall ELECTRICALLY NEUTRAL but carries both + and - charges.

At isoelectric point (pI), the amino acid exists fully as zwitter ion.

NCERT (NEP 2020): Ch.10 — Glucose p.293 | Anomers p.291-292 | Protein structures p.305-310 | Zwitter ion p.302-303

Q31(A). Alcohols, Phenols & Ethers — 5 sub-parts

(i) Symmetrical vs Unsymmetrical ethers:

Symmetrical (same groups on both sides of O):

$\text{CH}_3\text{-O-CH}_3$ (dimethyl ether / methoxymethane)

$\text{C}_2\text{H}_5\text{-O-C}_2\text{H}_5$ (diethyl ether / ethoxyethane)

Unsymmetrical (different groups on both sides of O):

$\text{CH}_3\text{-O-C}_2\text{H}_5$ (methoxyethane)

$\text{C}_2\text{H}_5\text{-O-C}_6\text{H}_5$ (ethoxybenzene)

(ii) Propene $\rightarrow$ Propan-1-ol (Hydroboration-Oxidation):

(Step 1) $\text{CH}_3\text{-CH=CH}_2 + \text{BH}_3 \rightarrow \text{CH}_3\text{-CH}_2\text{-CH}_2\text{-B}$ (trialkylborane)

B adds to LESS substituted C (terminal C1)

(Step 2) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-B} + \text{H}_2\text{O}_2/\text{NaOH} \rightarrow \text{CH}_3\text{-CH}_2\text{-CH}_2\text{-OH}$

(propan-1-ol)

Anti-Markovnikov: OH goes to terminal carbon.

(iii) Ethanal $\rightarrow$ Propan-2-ol (Grignard reaction):

Ethanal structure:

$\text{CH}_3\text{-CHO}$ (aldehyde, C2)

(Step 1) $\text{CH}_3\text{-CHO} + \text{CH}_3\text{MgBr} \xrightarrow{\text{dry ether}} \text{CH}_3\text{-CH(OMgBr)-CH}_3$

(Step 2) $\text{CH}_3\text{-CH(OMgBr)-CH}_3 + \text{dil. HCl/H}_2\text{O} \rightarrow \text{CH}_3\text{-CHOH-CH}_3$

(propan-2-ol, secondary alcohol)

CH_3MgBr (Grignard) adds to C=O . Hydrolysis gives 2 deg alcohol.

(iv) bp: Ethanol > Methoxymethane — Justify:

Ethanol ($\text{C}_2\text{H}_5\text{OH}$, bp = 78 deg C):

$\text{C}_2\text{H}_5\text{-O-H} \cdots \text{H-O-C}_2\text{H}_5$ (intermolecular H-bonds via -OH group)

Methoxymethane ($\text{CH}_3\text{-O-CH}_3$, bp = -23 deg C):

No O-H bond $\rightarrow$ CANNOT form H-bonds with itself

Only weak van der Waals (dipole-dipole) forces

H-bonds in ethanol are much stronger $\rightarrow$ needs more energy to break $\rightarrow$ MUCH higher bp.

(v) Why pyridine is used in acylation of alcohols:

$\text{R-OH} + \text{CH}_3\text{COCl} \rightarrow \text{R-O-CO-CH}_3$ (ester) + HCl

HCl is produced as byproduct during the reaction.

$\text{HCl} + \text{C}_5\text{H}_5\text{N}$ (pyridine) $\rightarrow \text{C}_5\text{H}_5\text{NH}^+\text{Cl}^-$ (pyridinium chloride)

Pyridine (base) neutralises HCl immediately:

1. Prevents acid-catalysed side reactions.

2. Drives esterification equilibrium forward (removing HCl product).

NCERT (NEP 2020): Ch.7 — Ethers p.238-239 | Hydroboration p.221 | Grignard p.222 | bp p.210-211 | Acylation p.228

Q31(B) OR. Prepare phenols from four sources

(i) From Chlorobenzene — DOW PROCESS:

(Step 1) $\text{C}_6\text{H}_5\text{-Cl} + \text{NaOH (aq)} \xrightarrow{[350 \text{ deg C, } 200 \text{ atm}]}$ $\text{C}_6\text{H}_5\text{-ONa}$ (sodium phenoxide)

(Step 2) $\text{C}_6\text{H}_5\text{-ONa} + \text{HCl} \rightarrow \text{C}_6\text{H}_5\text{-OH (phenol)} + \text{NaCl}$

High temperature and pressure required.

(ii) From Benzene sulphonic acid:

(Step 1) $\text{C}_6\text{H}_5\text{-SO}_3\text{H} + \text{NaOH} \xrightarrow{[\text{fused, } 300 \text{ deg C}]}$ $\text{C}_6\text{H}_5\text{-ONa} + \text{NaHSO}_3$

(Step 2) $\text{C}_6\text{H}_5\text{-ONa} + \text{HCl} \rightarrow \text{C}_6\text{H}_5\text{-OH} + \text{NaCl}$

(iii) From Benzene diazonium salt:

(Step 1) $\text{C}_6\text{H}_5\text{-NH}_2 + \text{NaNO}_2 + \text{HCl} \xrightarrow{[0-5 \text{ deg C}]}$ $\text{C}_6\text{H}_5\text{-N}_2^{+}\text{Cl}^{-}$ (diazonium salt)

(Step 2) $\text{C}_6\text{H}_5\text{-N}_2^{+}\text{Cl}^{-} + \text{H}_2\text{O} \xrightarrow{[50-60 \text{ deg C}]}$ $\text{C}_6\text{H}_5\text{-OH} + \text{N}_2(\text{gas}) + \text{HCl}$

(iv) From Cumene — INDUSTRIAL METHOD (most important):

Cumene structure: $\text{C}_6\text{H}_5\text{-CH}(\text{CH}_3)_2$ (isopropylbenzene)

(Step 1) $\text{C}_6\text{H}_5\text{-CH}(\text{CH}_3)_2 + \text{O}_2 \text{ (air)} \rightarrow \text{C}_6\text{H}_5\text{-C}(\text{CH}_3)_2\text{-O-OH}$
(cumene hydroperoxide)

(Step 2) Cumene hydroperoxide + dil. $\text{H}_2\text{SO}_4 \rightarrow \text{C}_6\text{H}_5\text{-OH} + \text{CH}_3\text{-CO-CH}_3$
(phenol) (acetone)

Gives TWO useful products: phenol AND acetone simultaneously.

NCERT (NEP 2020): Ch.7, p.218-220 — Preparation of phenols: all four methods

Q32(A). Boiling point & solution deviations — 5 sub-parts

(i) Why bp increases with non-volatile solute:

Non-volatile solute lowers vapour pressure (Raoult's law):

$$p_{\text{solution}} = x_{\text{solvent}} \times p^*_{\text{solvent}} \text{ (lower than pure solvent)}$$

For boiling: vapour pressure must = atmospheric pressure (1 atm).

Since vapour pressure is reduced, solution must be HEATED MORE to reach 1 atm.

$$\Delta T_b = K_b \times m \text{ (} K_b = \text{ebullioscopic constant, } m = \text{molality)}$$

(ii) Effect when solute dissociates:

$\text{NaCl} \rightarrow \text{Na}^{+} + \text{Cl}^{-}$ $i = 2$ (van't Hoff factor)

$$\Delta T_b = i \times K_b \times m \text{ (more particles = more elevation)}$$

(iii) Solute shows higher molecular mass than expected:

Higher apparent M = FEWER particles than expected = ASSOCIATION of solute molecules.

Example — acetic acid in benzene:

$2 \text{ CH}_3\text{COOH} \rightleftharpoons (\text{CH}_3\text{COOH})_2$ (dimer via H-bonds between two $-\text{COOH}$ groups)

Fewer particles $\rightarrow$ smaller colligative effect $\rightarrow$ higher observed molecular mass.

(iv) Reason for strong deviation from Raoult's law:

Positive deviation ($p > \text{ideal}$):

A-B interactions WEAKER than A-A and B-B

Example: ethanol + water (but water-ethanol is actually negative...)

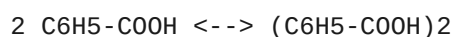
Better example: ethanol + cyclohexane (A-B weaker) $\rightarrow$ vapour pressure higher

Negative deviation ($p < \text{ideal}$):

A-B interactions STRONGER than A-A and B-B

Example: acetone + chloroform

$\text{CH}_3\text{-CO} + \text{H-CCl}_3$ forms H-bond $\rightarrow$ molecules held tightly $\rightarrow$ lower vapour pressure

(v) Van't Hoff factor < 1 for benzoic acid in benzene:**Dimerisation of benzoic acid in non-polar benzene:**

Two benzoic acid molecules form dimer via two O-H...O hydrogen bonds between -COOH groups

2 molecules $\rightarrow$ 1 dimer = FEWER particles

$i = \text{observed colligative property} / \text{calculated colligative property} < 1$

NCERT (NEP 2020): Ch.1 — Raoult's law p.8-11 | Non-ideal solutions p.11-14 | Colligative properties p.13-23 | Van't Hoff p.24-27

Q32(B) OR. Vapour pressure numerical + mole fraction definition**Numerical — Vapour pressure of glucose solution:****Given:**

p^* (pure water at 293K) = 18 mmHg

Mass of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) = 50 g

Mass of water = 250 g

Molar mass of glucose:

$$\text{C}_6\text{H}_{12}\text{O}_6 = 6(12) + 12(1) + 6(16) = 72 + 12 + 96 = 180 \text{ g/mol}$$

Calculations:

$$n(\text{glucose}) = 50 / 180 = 0.278 \text{ mol}$$

$$n(\text{water}) = 250 / 18 = 13.89 \text{ mol}$$

$$x(\text{water}) = 13.89 / (13.89 + 0.278) = 13.89 / 14.168 = 0.9804$$

Apply Raoult's law:

$$p(\text{solution}) = x(\text{water}) \times p^*(\text{water})$$

$$p(\text{solution}) = 0.9804 \times 18 = 17.65 \text{ mmHg}$$

Lowering of vapour pressure:

$$\Delta p = p^* - p = 18 - 17.65 = 0.35 \text{ mmHg}$$

ANSWER: Vapour pressure = 17.65 mmHg

Mole fraction — Definition:

Mole fraction of component 1 in a binary solution:

$$x_1 = n_1 / (n_1 + n_2)$$

n_1 = moles of component 1

n_2 = moles of component 2

$x_1 + x_2 = 1$ (sum of all mole fractions = 1)

NCERT (NEP 2020): Ch.1 — Mole fraction p.2-3 | Raoult's law vapour pressure p.8-10 | Worked examples p.11-13

Q33(A). Lucas test (3M) + Dehydration mechanism (2M)

LUCAS TEST — Distinguish 1 deg, 2 deg, 3 deg alcohols:

Reagent: Anhydrous ZnCl_2 + Conc. HCl = Lucas reagent

Observation: Turbidity = insoluble alkyl chloride formed in solution

3 degree alcohol — IMMEDIATE turbidity:



(tert-butanol) (tert-butyl chloride, INSOLUBLE)

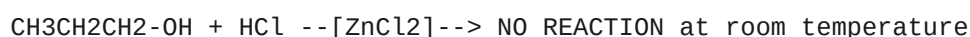
Stable $(\text{CH}_3)_3\text{C}(+)$ carbocation forms INSTANTLY $\rightarrow$ fast $\text{S}_\text{N}1$ $\rightarrow$ immediate turbidity

2 degree alcohol — turbidity in ~5 min:



2 deg carbocation less stable $\rightarrow$ slower $\text{S}_\text{N}1$ $\rightarrow$ turbidity after 5 min

1 degree alcohol — NO turbidity at room temp:

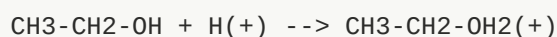


1 deg carbocation too unstable $\rightarrow$ $\text{S}_\text{N}1$ does not occur at room temp

MECHANISM OF DEHYDRATION OF ETHANOL (acid-catalysed E_1):

Conditions: Conc. H_2SO_4 , 170 deg C

Step 1 — Protonation of -OH (fast, reversible):



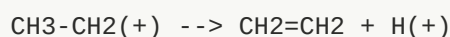
(oxonium ion / protonated alcohol)

Step 2 — Loss of water / carbocation formation (slow = rate determining step):



(ethyl carbocation)

Step 3 — Loss of beta-H / alkene formation (fast):



(ethylene) (H^+ regenerated = catalyst)

At 140 deg C: $\text{S}_\text{N}2$ gives diethyl ether. At 170 deg C: E_1 gives ethylene.

NCERT (NEP 2020): Ch.7 — Lucas test p.228 | Dehydration mechanism p.232-234

Q33(B) OR. $\text{S}_\text{N}1$ — five sub-parts

(i) Why iodide is a better leaving group than chloride:

!Bond strengths comparison:

C-F bond: 485 kJ/mol (strongest) $\rightarrow$ worst leaving group

C-Cl bond: 328 kJ/mol

C-Br bond: 276 kJ/mol

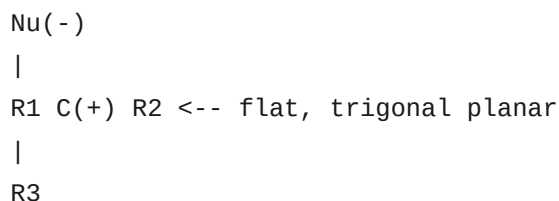
C-I bond: 240 kJ/mol (weakest) $\rightarrow$ best leaving group

I⁻ is larger, more polarisable, weaker base $\rightarrow$ more stable anion when it leaves.

Leaving group ability: $I(-) > Br(-) > Cl(-) > F(-)$

(ii) Why SN1 leads to racemisation:

Carbocation intermediate is PLANAR (sp² hybridised):



Nu(-) can attack from TOP (front) OR BOTTOM (back) with equal probability
Result: 50% R + 50% S enantiomers = RACEMIC MIXTURE = zero optical rotation

(iii) Why tertiary halides prefer SN1:

Carbocation stability (hyperconjugation + inductive effect):

3 deg: (CH₃)₃C(+) -- three alkyl groups donate electrons --> MOST STABLE

2 deg: (CH₃)₂CH(+) -- two alkyl groups --> moderately stable

1 deg: CH₃CH₂(+) -- one alkyl group --> unstable

methyl: CH₃(+) -- no alkyl group --> LEAST STABLE

3 deg carbocation forms EASILY --> fast SN1.

Also: 3 deg halides are too bulky for SN2 backside attack.

(iv) Racemisation — Definition:

Conversion of an optically active compound (pure enantiomer, [α] ≠ 0)

into an equimolar mixture of both enantiomers (racemic mixture, [α] = 0).

50% R-form + 50% S-form = racemic mixture = optically inactive

(v) Zaitsev's Rule (Saytzeff's Rule):

Statement: In elimination, major product is the MORE SUBSTITUTED alkene.

Example — dehydrohalogenation of 2-bromobutane:



Major: CH₃-CH=CH-CH₃ (but-2-ene, more substituted)

Minor: CH₃-CH₂-CH=CH₂ (but-1-ene, less substituted)

H is removed from beta-C having fewer H atoms (more substituted alkene = more stable).

NCERT (NEP 2020): Ch.6 — SN1 mechanism p.184-191 | Leaving groups p.183 | Racemisation p.190-191 | Zaitsev p.197